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SOUR-TASTE TOLERANCE IN FOUR SPECIES OF NONHUMAN PRIMATES

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Abstract—The taste of most fruits is characterized by a mixture of sensations termed sweet and sour by humans, and the food selection behavior of primates suggests that they may use the relative salience of sweetness and sourness to assess palatability of potential food items. Therefore, taste responses of six squirrel monkeys, five pigtail macaques, four olive baboons, and four spider monkeys to sweet–sour taste mixtures were assessed in two-bottle preference tests of brief duration (2 min). Monkeys were given the choice between a reference solution of 50 mM sucrose and mixtures containing 10, 30, or 50 mM citric acid plus 10, 20, 50, 100, 200, 400, 800, or 1000 mM sucrose. We found that the four species differed markedly in their acceptance of physiological concentrations of sour-tasting citric acid. Whereas olive baboons showed the highest degree of sour-taste tolerance and actually preferred most of the sweet–sour taste mixtures over sweet-tasting reference solutions, squirrel monkeys showed the lowest degree of sour-taste tolerance and rejected most of the sweet–sour taste mixtures even when they contained considerably more sucrose than the reference solutions. Additional tests demonstrated that the preference for sweet–sour taste mixtures was not based on masking effects. Rather, the animals perceived both the sweetness and the sourness of the taste mixtures and made a trade-off between the attractive and aversive properties of the two taste qualities. The results of this study suggest that the proximate reason for the marked differences in acceptance of sweet–sour taste mixtures are differences among species in the hedonic evaluation of the sour taste of citric acid. Possible ultimate reasons, which do not necessarily exclude, but may complement each other, include evolutionary

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adaptation to dietary specialization, avoidance of competition pressure, and phylogenetic relatedness.

Key Words—Sour-taste tolerance, sweet–sour taste mixtures, *Saimiri, Ateles, Macaca, Papio*, food preference.

INTRODUCTION

The sense of taste has been investigated anatomically (e.g., Pritchard, 1991; Rolls, 1995), behaviorally (e.g., Glaser, 1986; Laska, 1996, 1999, 2000), and electrophysiologically (e.g., Plata-Salaman et al., 1993, 1995; Hellekant and Roberts, 1995; Scott and Plata-Salaman, 1999) in a number of nonhuman primate species. However, most studies so far have concentrated on detectability of the four basic taste qualities, usually using sucrose, sodium chloride, hydrochloric acid, and quinine hydrochloride as the only, prototypic stimuli, or have assessed the properties of artificial tastants such as high-potency sweeteners or taste modifiers (e.g., Glaser, 1993; Nofre et al., 1996). There is sparse information as to taste perception of nonhuman primates for heterogenous taste mixtures (Plata-Salaman et al., 1996). This is remarkable given that the taste of almost any food is composed of more than one taste quality (Galef, 1996). The taste of most fruits, for example, is characterized by a mixture of sensations termed sweet and sour by humans (Nagy and Shaw, 1980), and the food selection behavior of primates suggests that they may use gustatory cues, and the relative salience of sweetness and sourness in particular, to assess palatability of a potential food item (Clutton-Brock, 1977; Laska, 2001). This appears to make sense in terms of optimal foraging (Stephens and Krebs, 1986), as the ratio between sour-tasting acids and sweet-tasting carbohydrates has been shown to change systematically as a function of the degree of ripeness in a large variety of fruits (Hulme, 1970) and thus is a good indicator of their nutritional value (Souci et al., 1989).

Animals that include fruits into their diet have to decide frequently whether a potential food item is already sweet enough to be consumed despite the presence of sourness, or whether it is still too sour to be consumed despite the presence of sweetness. Thus, it was the aim of this study to assess sour-taste tolerance in four species of nonhuman primates. To this end, we employed two-bottle preference tests of brief duration (2 min). This method makes it possible to directly measure preferences for or aversions to tastants, and largely rules out the influence of postingestive factors on the animal's ingestive behavior. More specifically, the monkeys were given a choice between a moderately sweet-tasting sucrose solution as a reference and mixtures containing variable concentrations of sour-tasting citric acid plus variable concentrations of sucrose.

It is important to note that, theoretically, such an approach could yield three different results: (A) The animal always prefers the solution containing no

sour-tasting citric acid, irrespective of the concentration of sucrose in the mixture. This would reflect an extremely low sour-taste tolerance. (B) The animal always prefers the solution containing the higher concentration of sucrose, irrespective of the concentration of citric acid in the mixture. This would reflect an extremely high sour-taste tolerance. (C) The animal displays a choice behavior that lies between the extremes outlined in (A) and (B), and shows preferences for one alternative or the other taking into consideration both the concentrations of the sour-tasting citric acid and the sweet-tasting sucrose.

The four primate species employed here, squirrel monkeys, spider monkeys, olive baboons, and pigtail macaques, are known to differ—at least to some degree—in their dietary habits (Clutton-Brock, 1977; Caldecott, 1986; Chapman, 1987; Ross, 1992) allowing us to address the question whether possible differences in sour-taste tolerance may reflect an evolutionary adaptation to dietary specialization. The possibility to include two New World primate species and two Old World primate species allowed us to additionally address the question whether the degree of phylogenetic relatedness rather than dietary specialization may affect the sour-taste tolerance of nonhuman primates.

METHODS AND MATERIALS

Animals. Testing was carried out using six male adult squirrel monkeys (*Saimiri sciureus*), two male and three female adult pigtail macaques (*Macaca nemestrina*), two male and two female adult olive baboons (*Papio hamadryas anubis*), and one male and three female adult spider monkeys (*Ateles geoffroyi*). Animals of all four species were housed as social groups in enclosures with adjacent single cages, which could be closed by sliding doors to allow temporary separation of animals for individual testing (for details of maintenance see Laska, 1996, 2000; Laska et al., 1996, 1999). Animals were fed commercial monkey chow, fresh fruit, and vegetables *ad libitum*, but were deprived of water overnight before testing on the following morning. The amount of food offered daily to the animals was such that leftovers were still present on the floor the next morning, and it was unlikely that ravenous appetite affected the animals' ingestive behavior in the tests.

Procedure. Sour-taste tolerance to three concentrations of citric acid was assessed by using a two-bottle preference test of short duration (Richter and Campbell, 1940). Tests were performed daily, and twice each day, approximately 2 and 1 hr before feeding, animals were separated and allowed 2 min to drink from a pair of simultaneously presented graduated cylinders (100 ml volume for *Saimiri* and *Ateles*, and 700 ml for *Papio* and *Macaca*) with metal drinking spouts.

In experiment 1, the monkeys were given a choice between a 50 mM sucrose solution and mixtures containing 10 mM citric acid plus variable concentrations

of sucrose (10, 20, 50, 100, 200, 400, 800, and 1000 mM). In experiments 2 and 3, the same combinations were tested except for the fact that the mixtures contained 30 and 50 mM citric acid, respectively, rather than 10 mM citric acid. The rationale for choosing these concentrations of citric acid was that they represent the range of concentrations present in a large variety of tropical fruits (Nagy and Shaw, 1980). Furthermore, they are detectable for all four primate species, as earlier studies have found taste preference thresholds for citric acid to be lower than 10 mM (Laska, 1999, 2000; Laska et al., 2000a,b; Laska et al., unpublished data). The rationale for choosing a 50 mM sucrose solution as the reference was that this concentration is also clearly detectable and attractive for all four primate species, but is only a factor of five (respectively, 10 for the spider monkeys) above their taste preference threshold (Laska, 1996, 2000; Laska et al., 1996, 1999) and, thus, represents a moderately sweet taste stimulus. In experiment 4, monkeys were given a choice between those mixtures of citric acid plus sucrose that were preferred over the 50 mM sucrose solution in experiments 1–3 and a sucrose solution containing the same concentration as present in the preferred mixture. This was done to get an indication of whether the preferences for the sweet–sour taste mixtures observed in experiments 1–3 were based on masking of the sour taste by the sweet taste or whether the animals still perceived the sourness and made a trade-off taking both taste qualities into consideration.

To keep up the animals' motivation and willingness to cooperate, testing did not follow a strict order of increasing or decreasing sucrose concentrations in the mixtures, but followed a pseudo-randomized scheme in which trials with taste mixtures containing high and, thus, presumably attractive concentrations of sucrose were alternated with mixtures containing low and, thus, presumably less attractive concentrations of sucrose. In all four experiments, each pair of stimuli was presented 10 times, and the position of the stimuli was pseudorandomized to counterbalance possible position preferences. All animals had served in previous studies using the same method (Laska, 1994, 1996, 1997, 1999, 2000; Laska et al., 1996, 1998, 1999b, 2000a,b), they were trained to enter the single cages voluntarily and were completely accustomed to the procedure.

The experiments reported here comply with the *Guide for the Care and Use of Laboratory Animals* (National Institutes of Health Publication No. 86-23, revised 1985) and also with current German and Mexican laws.

Data Analysis. For each animal, the amount of liquid consumed from each bottle was recorded, summed for the 10 test trials with a given stimulus combination, converted to percentages (relative to the total amount of liquid consumed from both bottles), and 66.7% (i.e., 2/3 of the total amount of liquid consumed) was taken as criterion of preference. This rather conservative criterion was chosen for purposes of comparability of data, as the same criterion had been used in previous studies using the same method with the same primate species (Laska, 1994, 1996, 1997, 1999, 2000; Laska et al., 1996, 1998,

1999a,b, 2000a,b, 2001), and to avoid misinterpretation of data due to a too liberal criterion.

Additionally, two-tailed binomial tests (Siegel and Castellan, 1988) were performed, and an animal was only regarded as preferring one of the two alternative stimuli if it reached the criterion of 66.7%, and consumed more from the bottle containing the preferred stimulus in at least 8 out of 10 trials (binomial test, $P < 0.05$).

Preliminary analysis of the data indicated that there were no reliable differences in choice behavior and liquid consumption between the male and the female subjects of a species and between the first and the second presentation of the day. Intraindividual variability in the amount of liquid consumed across the 10 test trials with a given stimulus combination was low and averaged less than 20%. Thus, a theoretically possible bias in the overall preference score due to excessive drinking in aberrant trials did not occur. Therefore, the data for the males and the females of a species obtained in the 10 test trials were combined and are reported as group means and standard deviations.

RESULTS

Figure 1 shows the mean taste responses of the four primate species given a choice between a 50 mM sucrose solution and mixtures of 10, 30, and 50 mM citric acid, respectively, plus various concentrations of sucrose. The squirrel monkeys needed a concentration of 200 and 800 mM sucrose to prefer mixtures with 10 and 30 mM citric acid, respectively, over a reference solution of 50 mM sucrose. With lower sucrose concentrations, *Saimiri sciureus* either rejected the sweet–sour mixtures (with 10 and 20 mM sucrose in a mixture with 10 mM citric acid, and with 10–100 mM sucrose in a mixture with 30 mM citric acid) or consumed approximately equal amounts from both alternatives (with 50 and 100 mM sucrose in a mixture with 10 mM citric acid, and with 200 and 400 mM sucrose in a mixture with 30 mM acetic acid). With 50 mM citric acid, even a concentration of 1000 mM sucrose was not sufficient for the squirrel monkeys to prefer a mixture over a reference solution of 50 mM sucrose. Sweet-sour mixtures containing up to 400 mM sucrose were rejected, and only mixtures with 800 and 1000 mM sucrose were about equally attractive for *Saimiri sciureus* as the reference solution.

The spider monkeys needed only a concentration of 100 mM sucrose to prefer sweet–sour taste mixtures with 10 and 30 mM citric acid over a sweet-tasting reference solution of 50 mM sucrose. With lower concentrations of sucrose in the mixture, *Ateles geoffroyi* generally rejected the sweet–sour mixtures, except for the mixture of 10 mM citric acid and 50 mM sucrose, which was consumed at approximatley equal amounts compared to the reference solution. With 50 mM citric acid, the spider monkeys needed a concentration of 800 mM sucrose to

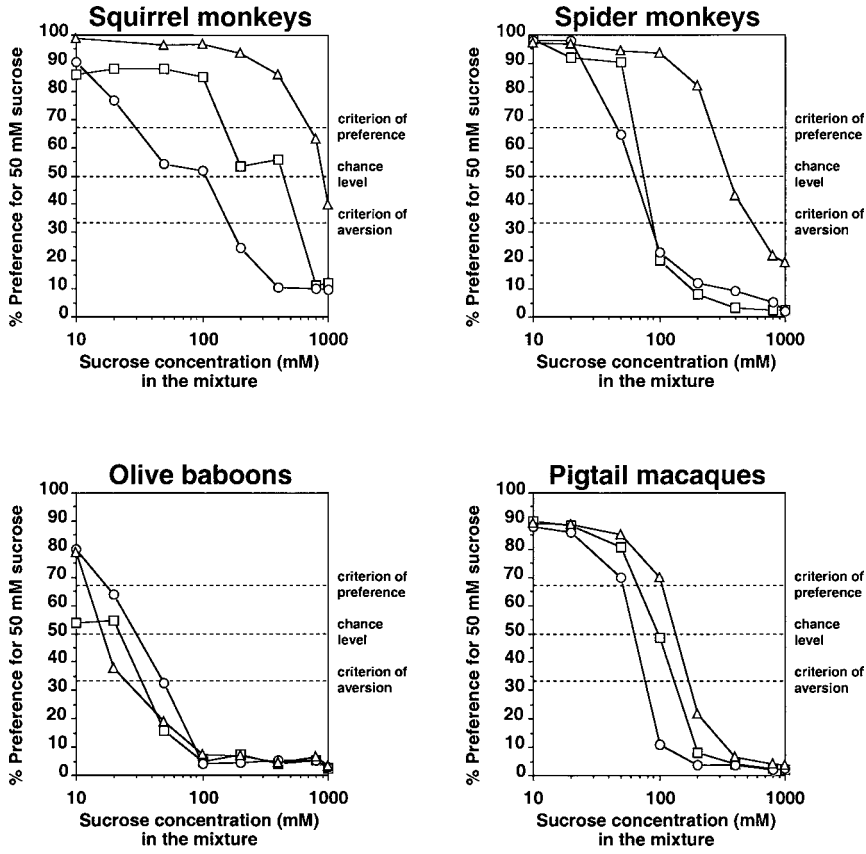


FIG. 1. Taste responses of six squirrel monkeys, four spider monkeys, four olive baboons, and five pigtail macaques given a choice between a 50 mM sucrose solution and mixtures of 10 mM (circles), 30 mM (squares), and 50 mM (triangles) citric acid plus various concentrations (10–1000 mM) of sucrose. Each data point represents the mean value of 10 test trials of 2 min per animal.

prefer a sweet–sour taste mixture over a sweet-tasting reference solution of 50 mM sucrose. A mixture containing 400 mM sucrose was as attractive for *Ateles geoffroyi* as the reference, and mixtures with 200 mM sucrose or less were rejected.

The olive baboons needed only a concentration of 50 mM sucrose to prefer sweet–sour taste mixtures with 10, 30, and 50 mM citric acid, respectively, over a sweet-tasting reference solution of 50 mM sucrose. Only with mixtures containing the lowest sucrose concentration of 10 mM plus 10 and 50 mM citric acid, respectively, *Papio hamadryas anubis* showed a preference for the reference solution.

TABLE 1. CONCENTRATIONS OF SUCROSE (MM) NEEDED IN MIXTURES WITH CITRIC ACID TO INDUCE A SIGNIFICANT PREFERENCE OVER A SWEET-TASTING REFERENCE SOLUTION OF 50 MM SUCROSE

	Concentration of citric acid (mM) in the mixture		
Species	10	30	50
Squirrel monkeys	200	800	>1000
Spider monkeys	100	100	800
Olive baboons	50	50	50
Pigtail macaques	100	200	200

Sweet-sour mixtures containing 20 mM sucrose were as attractive as the 50 mM sucrose reference at all three concentrations of citric acid tested.

The pigtail macaques needed a concentration of 100 mM sucrose to prefer a sweet–sour taste mixture with 10 mM citric acid over a sweet-tasting reference solution of 50 mM sucrose. With 30 and 50 mM citric acid, *Macaca nemestrina* needed 200 mM sucrose to show a preference for the mixtures when a 50 mM sucrose solution was presented as the alternative. Sweet–sour mixtures containing 10–50 mM sucrose were rejected at all three concentrations of citric acid tested. A mixture of 100 mM sucrose plus 30 mM citric acid was as attractive for the pigtail macaques as the reference solution.

With all four species, interindividual variability in the percentage of preference for a given solution was generally low. Standard deviations across all test combinations averaged 6.8%, and in the majority of cases were smaller than 5%.

Table 1 summarizes the lowest concentrations of sucrose needed in mixtures with citric acid to induce a preference over a moderately sweet-tasting reference solution of 50 mM sucrose. Marked differences among the species tested are apparent. Whereas the olive baboons needed the lowest sucrose concentrations to make mixtures with citric acid more attractive than the reference solution, the squirrel monkeys needed the highest sucrose concentrations. More specifically, olive baboons actually preferred a sweet–sour mixture of citric acid plus sucrose over a sweet solution containing the same concentration of sucrose only. Squirrel monkeys, on the other hand, needed sucrose concentrations that were 4- and 16-fold higher than that of the sweet-tasting reference solution to make sweet–sour mixtures with 10 and 30 mM citric acid attractive, and with a mixture containing 50 mM citric acid even a sucrose concentration 20 times that of the reference solution was not sufficient to make it preferable over the reference. The pigtail macaques and spider monkeys fell in between these extremes and needed sucrose concentrations 2- to 4-fold (*Macaca nemestrina*) and 2- to 16-fold (*Ateles geoffroyi*)

higher than that of the sweet-tasting reference solution to show a preference for sweet–sour taste mixtures containing 10, 30, and 50 mM citric acid, respectively.

To get an indication of whether these preferences for sweet–sour taste mixtures were based on masking of the sour taste by the sweet taste or whether the animals still perceived the sourness and made a trade-off taking both taste qualities into consideration, an additional experiment was performed. Figure 2 shows the mean taste responses of the four primate species given a choice between mixtures of citric acid plus sucrose that were preferred over the 50 mM sucrose solution in the experiments described above, and a sucrose solution containing the same concentration as present in the preferred mixture. In all 11 cases (the 12th case, a

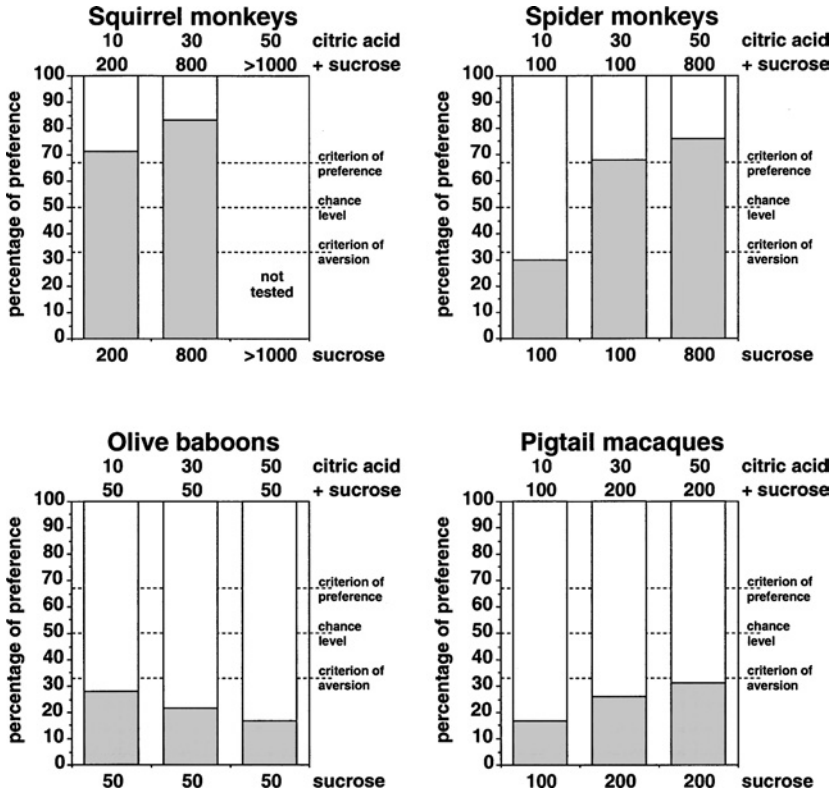


FIG. 2. Taste responses of six squirrel monkeys, four spider monkeys, four olive baboons, and five pigtail macaques given a choice between mixtures of citric acid plus sucrose that were preferred over the 50 mM sucrose solution in experiments 1–3, and a sucrose solution containing the same concentration as present in the preferred mixture. Each bar represents the mean value of 10 test trials of 2 min per animal.

mixture containing 50 mM citric acid with the squirrel monkeys was not tested as the animals failed to show a preference for any of the mixtures tested, cf. Figure 1), the animals showed marked preferences for one of the alternatives. Whereas the olive baboons and the pigtail macaques preferred the sweet–sour mixtures over a sweet-tasting alternative containing the same concentration of sucrose as present in the mixture, the squirrel monkeys, in contrast, preferred the sweet-tasting solution over the sweet–sour mixtures. Spider monkeys showed a preference for the sweet–sour mixture when the concentration of citric acid was low (10 mM), and rejected it when the concentration was medium or high (30 and 50 mM, respectively). These results indicate that all four primate species tested perceived the citric acid in the mixtures and, thus, suggest that the preferences shown in experiments 1–3 were not based on masking of the sour taste by the sweet taste, but rather on a trade-off between the attractive and aversive properties of the two taste qualities.

DISCUSSION

The results demonstrate that squirrel monkeys, spider monkeys, olive baboons, and pigtail macaques differ markedly in their tolerance to physiological concentrations of sour-tasting citric acid. Whereas olive baboons showed the highest degree of sour-taste tolerance among the four species and actually preferred most of the sweet–sour taste mixtures over sweet-tasting reference solutions, squirrel monkeys showed the lowest degree of sour-taste tolerance and rejected most of the sweet–sour taste mixtures even when they contained considerably more sucrose than the reference solutions. Additional tests demonstrated that the preference for sweet–sour taste mixtures shown by all four species was not based on masking effects. Rather, the animals perceived both the sweetness and the sourness of the preferred taste mixtures and made a trade-off between the attractive and aversive properties of the two taste qualities.

What are the possible reasons that underlie the observed differences in sour-taste tolerance among the primate species tested? The possibility that differences in sensitivity for citric acid and/or sucrose may account for the differing acceptance of sweet–sour taste mixtures found here can largely be ruled out, as all four species tested have been shown to share the same taste preference threshold value for citric acid (Laska, 1999, 2000; Laska et al., 2000a,b) and three out of four species have been reported to share the same taste preference threshold value for sucrose (Laska, 1996, 2000; Laska et al., 1999a), with the fourth species (*Ateles geoffroyi*) displaying a value that is only a factor of 3 lower (Laska et al., 1996).

It is well established that masking effects, i.e., complete suppression of one taste by another, qualitatively different taste, or other types of qualitative

or quantitave interactions such as enhancement or blending may occur when using heterogenous taste mixtures (Schifferstein and Frijters, 1993; Halpern, 1997). Although the possibility that species differences in such interactions between the tastes of citric acid and sucrose might have been responsible for the differing sour-taste tolerance cannot be ruled out completely, this seems unlikely as there is no reason to believe that the primate species tested here differ fundamentally from each other with regard to possible taste interactions such as masking. In a comprehensive human psychophysical study on detection of tastes in heterogenous binary mixtures, Stevens (1996) showed that sucrose concentrations of a factor of 10 or 100, respectively, above threshold, elevated the detection threshold for citric acid in a mixture only by a factor of 2 or 4, respectively. Further, he reported that tastants termed sweet and sour by humans mask each other much less efficiently compared to other combinations of tastants. This latter finding is supported by findings from electrophysiological recordings in the gustatory cortex of *Macaca fascicularis*, which have shown that in binary taste mixtures sourness induced by HCl is suppressed least among the four basic taste qualities (Plata-Salaman et al., 1996). The authors concluded that sour-best cells in primates are less susceptible to mixture suppression than other taste cell subtypes. Thus, although interaction effects are likely to have occurred with some of the taste mixtures employed in this study, their extent is unlikely to explain the marked differences among species that we found.

Possible differences in the hedonic evaluation (evaluation of the dimension "pleasant–unpleasant") of the sweet taste of sucrose can also be excluded as an explanation as all four species have been shown to display preferences of similar strength for suprathreshold concentrations of sucrose (Laska, 1996, 2000; Laska et al., 1996, 1999a). The most likely explanation for our finding is that the species differed in their hedonic evaluation of the sour taste of citric acid. This supposition is supported by the finding that squirrel monkeys generally reject suprathreshold concentrations of citric acid (Laska, 1999), whereas spider monkeys and pigtail macaques prefer low, but detectable, concentrations of this acid and reject only concentrations $\geq$20 and 50 mM, respectively (Laska, 2000; Laska et al., 2000a,b), and olive baboons even prefer concentrations up to 200 mM of this sour tastant (Laska, unpublished data).

Human and nonhuman primates have repeatedly been shown to be capable of learning to accept and even prefer tastes that they initially reject (Rozin and Schiller, 1980; Rozin and Kennel, 1983). Nevertheless, it is commonly agreed that hedonic responses of primates to tastants are generally innate and the result of an evolutionary adaptation to a species' dietary habits (Steiner et al., 2001).

Differences in dietary habits have repeatedly explained differences in taste performance among species (Kare, 1971; Pfaffmann, 1977; Spector, 2000). Among New World primates, for example, the degree of frugivory correlates positively with sensitivity for food-associated mono- and disaccharides (Laska, 1996; Laska et al.,

1996). Although considerable information on the dietary habits of the four primate species used in the present study are available (Caldecott, 1986; Chapman, 1987; Clutton-Brock, 1977; Ross, 1992), no definitive conclusions can be drawn whether differences in feeding specializations may account for the observed differences in sour-taste tolerance as, unfortunately, only few studies reported the proportion of unripe versus ripe fruits in their diet. It is interesting to note, however, that squirrel monkeys—the species showing the lowest degree of acceptance for sweet–sour taste mixture in this study—have also been reported to include the lowest proportion of fruits and seeds in their diet (26% of total intake) among the four species tested, whereas olive baboons (63%), pigtail macaques (72%), and spider monkeys (80%) include considerably higher proportions of fruit (Clutton-Brock, 1977).

An alternative explanation for the observed differences may be that preferring or at least accepting levels of acidic tastants that other species reject could help to avoid competition between sympatric frugivores for carbohydrate-rich fruits or might even give a species an advantage over competitors in the quest for fruits that are nutritionally valuable but still unripe and low in abundance (Gallina and Ausman, 1979). In their natural habitat, pigtail macaques compete for food with white-handed gibbons *Hylobates lar* and with Indonesian leaf monkeys *Presbytis potenziani* (Rowe, 1996). Whereas the latter two species do not include unripe fruits into their diets, pigtail macaques readily consume unripe *Ficus* fruits (Caldecott, 1986). As it is difficult to quantify competition pressure between species on a comparative basis, this explanation needs further verification.

Old World primates and New World primates differ in their ability to perceive substances such as aspartame and thaumatin, which both taste sweet to humans, suggesting that phylogenetic relatedness might also account for differences or correspondences in taste perception among species (Glaser, 1993). Indeed, in the present study, olive baboons and pigtail macaques, Old World primate species, accepted sweet–sour taste mixtures more readily than did squirrel monkeys and spider monkeys, New World primate species. Thus, the possibility of phylogenetic relatednes as an explanation for sour-taste tolerance also warrants further investigation.

In summary, the results of this study suggest that the major reason for the marked differences in acceptance of sweet–sour taste mixtures is that the primate species differed in their hedonic evaluation of the sour taste of citric acid. Other explanations that do not necessarily exclude, but may complement each other, include evolutionary adaptation to dietary specialization, avoidance of competition pressure, and phylogenetic relatedness.

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